

Foundations

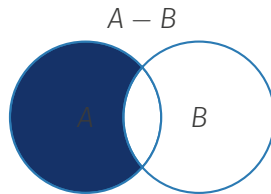
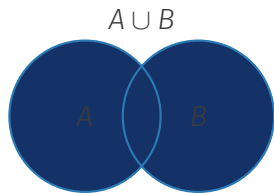
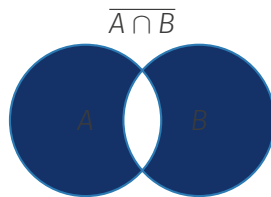
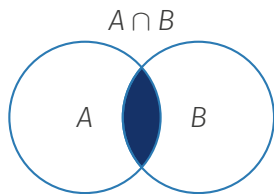
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Tuples vs. Sets

Tuple	Set
Parentheses as delimiters: (a, b, c) . Angle brackets as delimiters: $\langle a, b, c \rangle$. No delimiters: a, b, c .	Braces as delimiters: $\{a, b, c\}$
Repetition of elements matters. $(a, b, b, c) \neq (a, b, c)$	Repetition of elements does not matter. $\{a, b, b, c\} = \{a, b, c\}$
Order of elements matters. $(a, b, c) \neq (c, a, b)$	Order of elements does not matter. $\{a, b, c\} = \{c, a, b\}$
Contains finite number of elements.	Can contain infinite number of elements.

Sets Operations



Cartesian Product

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example

$$S = \{1, 2\}$$

$$X = \{x, y, z\}$$

$$S \times X = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$$

		X		
		x	y	z
S	1	(1, x)	(1, y)	(1, z)
	2	(2, x)	(2, y)	(2, z)

Relation

Relation

An n -place relation on a collection of sets

$$S_1, S_2, \dots, S_n$$

is a set

$$R \subseteq S_1 \times S_2 \times \dots \times S_n$$

of tuples of elements from S_1 through S_n .

We say that the elements $s_1 \in S_1$ through $s_n \in S_n$ are related by R if $(s_1, \dots, s_n) \in R$.

Notation

Often, we use fancy math symbols for an infix/mixfix notation:

$$a R b \equiv (a, b) \in R$$

$$a \not R b \equiv (a, b) \notin R$$

$$a \rightarrow b \equiv (a, b) \in \rightarrow$$

$$a \nrightarrow b \equiv (a, b) \notin \rightarrow$$

$$\Gamma \vdash e : t \equiv (\Gamma, e, t) \in \vdash:$$

Rules of Inference

$$\frac{\text{Premise}_1 \quad \text{Premise}_2 \quad \dots \quad \text{Premise}_n}{\text{Conclusion}}$$

Read:

If “Premise₁”, and “Premise₂”, ..., and “Premise_n” hold, we conclude “Conclusion”.

- Rules without premises are also called **axioms**.
- Rules of inference define **relations** in a **declarative** way.
- Plug several rules together like LEGO bricks to form a **derivation tree** that eventually ends on **axioms** to form a proof.

Example

$$\text{NULL} \frac{}{0 \text{ Nat}}$$

$$\text{Succ} \frac{n \text{ Nat}}{S(n) \text{ Nat}}$$

Example

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Does $S(S(S(0))) \text{ Nat}$ hold? (Is 3 a natural number?)

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Relation: Common Properties

Let R be a binary relation over $A \times A$.

R is ...

- ...**reflexive** if $\forall a \in A : a R a$.
- ...**symmetric** if $\forall a, b \in A : \text{if } a R b \Rightarrow b R a$.
- ...**antisymmetric** if $\forall a, b \in A : \text{if } a R b \text{ with } a \neq b \Rightarrow b \not R a$.
or equivalently: if $\forall a, b \in A : \text{if } a R b \text{ and } b R a \Rightarrow a = b$.

Closure

Transitive Closure

Let R be a binary relation. Then, R^+ is its **transitive closure**:

$$\text{ID} \frac{a R b}{a R^+ b}$$

$$\text{TRANS} \frac{a R^+ b \quad b R^+ c}{a R^+ c}$$

Reflexive Transitive Closure

Let R be a binary relation. Then, R^* is its **reflexive transitive closure**:

$$\text{ID} \frac{a R b}{a R^* b}$$

$$\text{TRANS} \frac{a R^* b \quad b R^* c}{a R^* c}$$

$$\text{REFL} \frac{}{a R^* a}$$

Example

$$\begin{array}{lcl} \text{NULL} \frac{}{0 \text{ Nat}} & \text{SUCC} \frac{n \text{ Nat}}{S(n) \text{ Nat}} & \text{PRED} \frac{n \text{ Nat}}{n \text{ pred } S(n)} \end{array} \quad \begin{array}{l} < \equiv \text{pred}^+ \\ \leq \equiv \text{pred}^* \end{array}$$

Does $S(0) \text{ pred } S(S(S(0)))$ hold? (Is $1 \leq 3$?)

TRANS $s(0) \leq s(s(s(0)))$

Example

$$\begin{array}{lcl} \text{NULL} \frac{}{0 \text{ Nat}} & \text{SUCC} \frac{n \text{ Nat}}{S(n) \text{ Nat}} & \text{PRED} \frac{n \text{ Nat}}{n \text{ pred } S(n)} \end{array} \quad \begin{array}{l} < \equiv \text{pred}^+ \\ \leq \equiv \text{pred}^* \end{array}$$

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Example

$$\begin{array}{lll}
 \text{NULL} \frac{}{0 \text{ Nat}} & \text{SUCC} \frac{n \text{ Nat}}{S(n) \text{ Nat}} & \text{PRED} \frac{n \text{ Nat}}{n \text{ pred } S(n)}
 \end{array}
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 \text{Succ} \frac{}{} \\
 \text{S(0) Nat} \\
 \text{PRED} \frac{}{} \\
 \text{S(0) pred S(S(0))} \\
 \text{ID} \frac{}{} \\
 \text{S(0) } \leq \text{ S(s(0))} \\
 \text{TRANS} \frac{}{} \\
 \text{S(0) } \leq \text{ S(S(S(0)))}
 \end{array}$$

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 \text{ID} \frac{}{S(0) \leq S(s(0))} \qquad \frac{}{S(S(0)) \leq S(S(S(0)))} \text{ID} \\
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Example

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 \end{array}
 \quad
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$$\begin{array}{c}
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\text{ID} \frac{\text{PRED} \frac{\text{Succ} \frac{\text{S(0) Nat}}{\text{S(s(0)) Nat}}}{\text{S(S(0)) pred S(S(S(0)))}}} {\text{S(S(0)) } \leq \text{ S(S(S(0)))}}} \\
\end{array}$$

Example

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$$\begin{array}{c}
\text{TRANS} \quad \frac{\text{ID} \quad \frac{\text{PRED} \quad \frac{\text{Succ} \quad \frac{\text{NULL} \quad \frac{0 \text{ Nat}}{\text{Succ}}}{\text{S}(0) \text{ Nat}}}{\text{S}(0) \text{ pred S}(S(0))}}{\text{S}(0) \leq \text{S}(s(0))}}{\text{S}(0) \leq \text{S}(S(S(0)))}} \\
\text{ID} \quad \frac{\text{PRED} \quad \frac{\text{Succ} \quad \frac{\text{S}(0) \text{ Nat}}{\text{S}(s(0)) \text{ Nat}}}{\text{S}(S(0)) \text{ pred S}(S(S(0)))}}{\text{S}(S(0)) \leq \text{S}(S(S(0)))}} \\
\text{PRED} \quad \frac{\text{Succ} \quad \frac{\text{S}(s(0)) \text{ Nat}}{\text{S}(S(S(0))) \text{ Nat}}}{\text{S}(S(S(0))) \text{ pred S}(S(S(S(0))))}} \\
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\text{NULL} \quad \frac{0 \text{ Nat}}{\text{S}(S(S(S(0)))) \text{ pred S}(S(S(S(S(0))))}}
\end{array}$$

Non-Binary Relations

Interpreting a 4-place relation

$$R \subseteq A \times B \times A \times B$$

as a set of pairs of pairs gives us technically a binary relation:

$$(a, b) R (a, b)$$

$$(a, b) R^+ (a, b)$$

$$(a, b) R^* (a, b)$$

We can think of a 3-place relation as a binary relation nested inside of another binary relation. For example, in the following 3-place relation

$$\sigma, e \rightarrow e$$

we build the reflexive, transitive closure of the binary relation $e \rightarrow e$ nested inside of $\sigma, e \rightarrow e$:

$$\sigma, e \rightarrow^* e$$

Further Notation: Functions

Let f be a function, then

- $\text{dom}(f)$ denotes the domain and
- $\text{codom}(f)$ the codomain of f .

We write $f[x \mapsto y]$ to express a function that is identical to f except that we add or update the mapping $x \mapsto y$:

$$f[x \mapsto y] : \text{dom}(f) \cup \{x\} \rightarrow \text{codom}(f) \cup \{y\}$$
$$x' \mapsto \begin{cases} y & \text{if } x' = x \\ f(x') & \text{otherwise.} \end{cases}$$

Further Notation: Sequence

- We write a sequence as $a_1, \dots, a_n =: \bar{a}$.
- ϵ denotes the empty sequence although it is sometimes just omitted.
- We write $a \in A$, if a occurs in the sequence A .
- We use $\bar{a} \in A$ as shorthand for $\forall a_i \in A : a_i \in A$.
- The comma operator performs concatenation. Thus, “ A, B ” concatenates two sequences A and B whereas “ A, a ” adds the element a to the sequence A .

Further Notation: Pairs, Sequence of Pairs

- We will often write $a : b$ to designate a pair if we want to stress that b is associated to a .
- We also view a sequence of pairs

$$A := a_1 : b_1, \dots, a_n : b_n =: \overline{a : b}$$

in which each a_i is unique as injective function:

$$A : \{a_1, \dots, a_n\} \rightarrow \{b_1, \dots, b_n\}$$

$$a_1 \mapsto b_1$$

...

$$a_n \mapsto b_n$$

Further Notation: Examples

$$\text{dom}(A) = \{a_1, \dots, a_n\}$$

$$\text{codom}(A) = \{b_1, \dots, b_n\}$$

$$A[x \mapsto y] = \overline{a : b}, x : y$$

$$A[a_1 \mapsto y] = a_1 : y, a_2 : b_2, \dots, a_n : b_n .$$